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Transfer learning for cross-scene 3D pavement crack detection based on enhanced deep edge features

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ABSTRACT

Inspired by heterogeneity of rapid-increasing 3D pavement data and the generalization ability of transfer learning, a robust and generalized framework for cross-scene 3D pavement-crack detection and attribute extraction was proposed in this paper, called profile component decomposition model with holistically nested edge detection (PCDM-HED). The core purpose of the PCDM-HED is to construct the enhanced deep edge features. By applying profile frequency, sparse characteristics of 3D profiles, and fusing the multiscale and multilevel edge characteristics of 3D depth maps in HED network, the robust enhanced feature can highlight the essential properties of cracks in heterogeneous 3D data. It overcomes the complex domain shifts caused by different 3D imaging conditions, pavement textures, and crack distributions in heterogeneous data. Cross-domain transfer experiments were carried out over seven 3D/2D datasets with 915 pavement sections. The results show that proposed PCDM-HED achieved average buffered Hausdorff scores of 90.17 to 96.42, recall scores of 0.84 to 0.91, and F-values of 0.85 to 0.89 in six different datasets without labeled samples. Compared with 9 groups of comparison results, including the traditional and related state-of-the-art methods, the transfer generalization effect of proposed PCDM-HED is more than 23% higher than that of comparison results. The proposed PCDM-HED makes full use of limited off-the-shelf samples, demonstrated strong transfer learning capability. It provides an effective solution for heterogeneous 3D pavement-crack detection tasks in engineering, in the case of limited labeled samples or even no corresponding labeled samples.

1. Introduction

Cracks are the most common pavement defects, and its automatic recognition is vital for pavement maintenance and management. Various pavement data acquisition systems have been developed and used in practical pavement-detection engineering applications. Among these, 3D laser scanning systems obtain depth information, overcome the influence of light and shadows (Qingquan et al., 2017; Fei et al., 2019; Zhang et al., 2018c; Du et al., 2021), and provide effective and comprehensive 3D databases for automatic pavement-crack detection (Hu et al., 2021; Weixing et al., 2019). Relevant 3D products, such as laser crack measurement systems (LCMS) (Qingquan et al., 2017; Laurent et al., 2012) and laser scanning profilers (LSP) (Zhang et al., 2018c; Gui et al., 2018) have been applied in real pavement-maintaining applications worldwide. With the widespread use of the engineering applications of these 3D systems, an increasing number of heterogeneous 3D pavement datasets have been created containing different texture backgrounds, crack types, and acquisition parameters. However, currently popular machine-learning methods, including deep-learning methods, are primarily built on homogeneous training and testing datasets. Thus, the automatic and rapid detection of pavement cracks in heterogeneous data requires practical applications.

Cracks usually exhibit obvious edge characteristics in 3D pavement data; however, owing to the complex background and inhomogeneity of cracks, improving the efficiency and effectiveness of crack detection still remains the main purpose of the current relevant methods (Gui et al., 2018). Typical 3D crack detection depends on edge detection or segmentation. Several recent algorithms have achieved excellent results for specific 3D pavement datasets. The relevant literature on 3D pavement-crack detection can be roughly divided into four categories based on data processing methods: (1) 3D data preprocessing 2D image crack detection (Lang et al., 2020; Li et al., 2017), (2) laser scanning profile processing (Gui et al., 2018; Sun et al., 2012), (3) 3D point cloud (Zhang et al., 2017a, 2018a), and (4) deep learning (Fei et al., 2019; Sholevar et al., 2022; Luo et al., 2022; Doğan and Ergen, 2022). First, preprocessing 2D image-based methods remove the cross-slope

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of vehicle vertical vibrations by designed preprocessing in 3D data; then, the 3D depth map is transferred to depth images, and the cracks are obtained by sound 2D crack detection methods. This category is primarily based on the fact that many cracks have edge characteristics in depth images and is the most common solution for 3D pavement-crack detection. These algorithms generally treat 3D pavement data as 2D pavement images only or largely use the 2D information contained in 3D data (Gui et al., 2019). Second, profile-based methods consider each profile as a basic unit considering the laser scanning data acquisition principle (Gui et al., 2018). Cracks in these elevation profiles show an obvious 'V' structure. The advantage of this category is that it can combine the acquisition principles of 3D profile data and analyze the characteristics of cracks using mature signal-processing models. However, these methods are typically ineffective for cracks oriented in the same direction as the profile. 3D model-based methods make full use of 3D information and consider the characteristics of cracks with different orientations. A 3D shadow model (Zhang et al., 2017a) and a 3D pavement reconstruction model (Zhang et al., 2018a) were designed to analyze crack characteristics. With the success of deep learning in recent years, neural networks have gradually become a research trend in crack-detection tasks (Weixing et al., 2019; Xu et al., 2021; Zhang et al., 2022; Wen et al., 2022a). However, existing supervised methods, including deep learning, generally deal with homogeneous data having the same characteristics and rely on a large number of labeled samples, and the labeling of cracks in 3D pavement data is an expensive and time-consuming task (Xu and Liu, 2022; Li et al., 2022; Gopalakrishnan et al., 2017).

Most above-mentioned algorithms have achieved satisfactory results for specific homogeneous datasets. However, there are strong heterogeneities in the 3D pavement data obtained from actual environments, such as differences caused by vehicle dynamic environments, texture backgrounds, and crack types. Based on limited heterogeneously labeled samples, it is difficult to achieve satisfactory results in heterogeneous 3D pavement-crack detection tasks in practical applications. Although unsupervised methods do not require training samples, they rely on a large number of adaptive thresholds to address different types of pavement cracks, and unsupervised algorithms are difficult to apply to heterogeneous data. Therefore, making maximum use of limited labeled samples to supervise and train more heterogeneous 3D pavement-crack detection tasks is a practical problem that must be solved (Xu and Liu, 2022; Li et al., 2022). Fortunately, transfer learning (TL) provides a solution for transferring labeled samples between heterogeneous yet related datasets (Roy, 2022). TL is more consistent with practical application requirements and can reuse valuable labeled samples to interpret newly obtained heterogeneous data and improve interpretation efficiency and model flexibility. In recent years, TL has been applied to 2D pavement-crack detection tasks and shown an attractive potential for real heterogeneous 3D crack detection. However, there are few studies on TL and domain-shift analysis for heterogeneous 3D pavement-crack detection.

Based on the current research status and practical problems in crack detection tasks, it is necessary to develop robust and generalized 3D crack detection methods with sample reuse and TL abilities for heterogeneous 3D data. The difficulty in applying TL to cross-domain 3D pavement-crack detection mainly includes three aspects: (1) there are more influencing factors between heterogeneous 3D data, such as driving postures, pavement deformations, pavement texture backgrounds, crack types, and depths; (2) it is difficult to directly use a TL model for 2D natural images in 3D pavement data; and (3) an unsupervised domain adaptation method is required because, in practice, there are usually no labeled samples for newly obtained 3D data. To solve these problems, this study first analyzed the domain-shift problems between heterogeneous 3D pavement data. Subsequently, a cross-domain 3D pavement-crack detection method was proposed based on the profile frequency, sparse characteristics, and multiscale and multilevel edge characteristics of cracks in 3D data. The main contributions and advantages of this study are as follows:

(1) The proposed robust cross-domain 3D pavement-crack recognition profile component decomposition model with holistically nested edge detection (PCDM-HED) with TL capability provides an effective and generalized solution for actual heterogeneous 3D and 2D crack detection tasks by reusing limited off-the-shelf samples and overcoming complex domain shifts.

(2) An enhanced deep edge feature was developed based on the profile frequency, sparse characteristics, and multiscale and multilevel edge characteristics. This feature mastered complex domain shifts in heterogeneous 3D pavement data caused by different 3D imaging conditions, pavement textures, and crack distributions.

(3) The PCDM-HED achieves average buffered Hausdorff scores of 90.17 to 96.42 in six different datasets without labeled samples, and demonstrates strong TL ability across different 3D pavement scenes. Compared with 9 groups of comparison results, including the traditional and related state-of-the-art methods, the transfer generalization effect of proposed PCDM-HED is more than 23% higher than that of comparison results. Heterogeneous 3D pavement datasets with different crack types and pavement textures from seven pavement-maintenance engineering projects were established and released.

The following summarizes the organization of the remainder of this study. Section 2 briefly describes the 3D pavement data acquisition system, domain shift problems between heterogeneous 3D pavements, and related works. Section 3 introduces the proposed methods for detecting cross-domain 3D pavement cracks. Section 4 introduces the experimental dataset, validates the algorithms, and analyzes the results, and Section 5 summarizes the work and limitations.

2. Domain shifts among heterogeneous 3D data and related works

2.1. 3D pavement data acquisition system

Here, a laser scanning profiler (LSP, ZOYON Ltd., Wuhan, China) was used for the 3D pavement data collection system (Qingquan et al., 2017; Zhang et al., 2018c; Gui et al., 2018) as shown in Fig. 1. Vehicle-carrying 3D laser scanning systems measured the elevation profiles of pavement based on the principle of triangulation. In practical applications, the pavement inspection speed must meet the normal driving speed range of 0–120 km/h. Two 3D cameras were combined to capture profiles with lane widths of approximately 4 m. The resolution in the transverse direction was approximately 1 mm with a profile sampling interval of 1 mm. The sampling interval can be adjusted from 1 to 5 mm along the traffic direction. The depth accuracy was 0.3 mm, and the depth measurement range was approximately 0.2 m. The typical 3D crack data for different pavements obtained using the applied system are shown in Fig. 1. The basic elevation profile obtained by each camera was 1 * 2048, and every 1000 profiles constituted a basic data unit. For the experimental data, the data size was 2048 * 1000.

2.2. Domain-shift problems among heterogeneous 3D pavement data

The 3D pavement contains mixed elevations of many components, including the elevations of vehicle vertical vibrations, pavement deformation, pavement texture, cracks, and other defects. Therefore, the crack characteristics were not evident in the original 3D data, and many small cracks are easily affected by pavement texture from the perspective of elevation. Owing to the complexity of the data acquisition environment and system, there were significant differences among the 3D pavement data obtained for different pavements, called cross-domain shift problems in TL. These differences and their impacts on heterogeneous 3D crack detection were as follows:

(1) The differences originated from different system parameters. The laser scanning system automatically collected 3D data in a dynamic driving environment. The system parameters were adjustable, and the cracks exhibited different depths and structures in different 3D pavement data. Thus, it was necessary to enhance the characteristics of

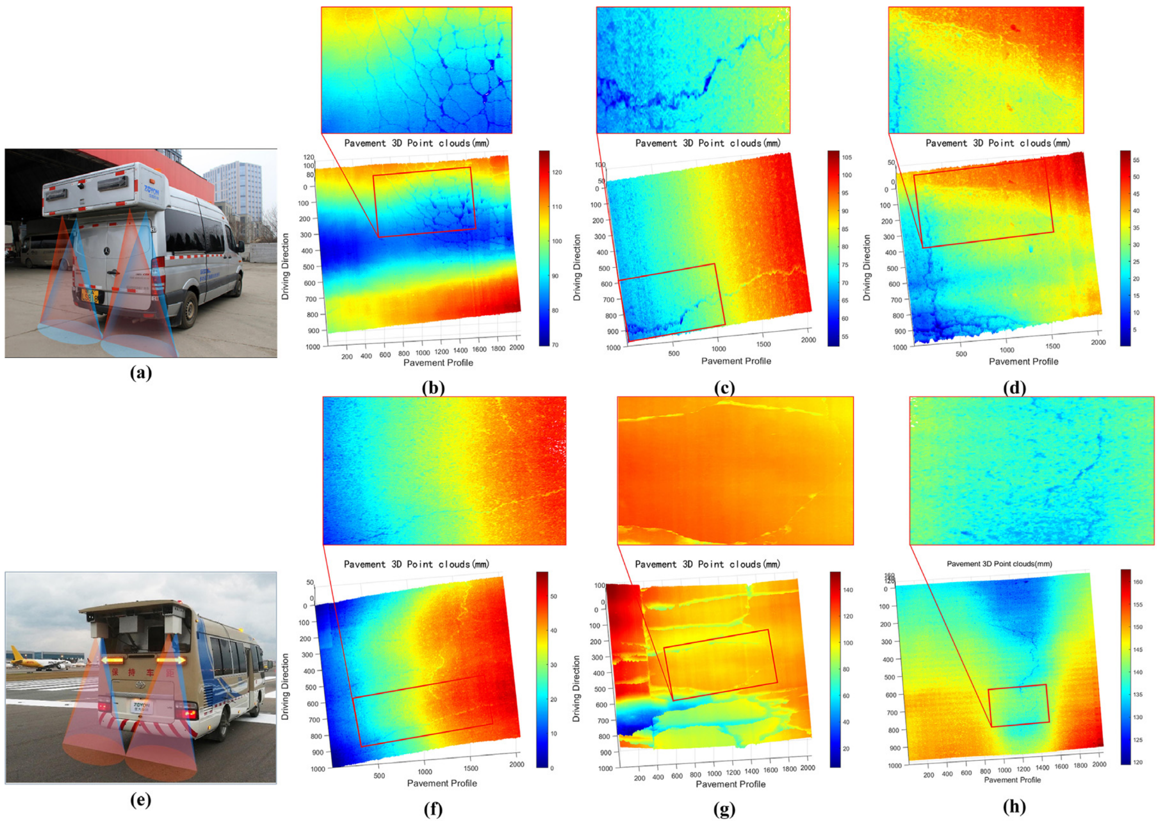

Fig. 1. Applied 3D pavement data collection system and typical 3D crack data in different pavements, (a)(e) typical vehicle-carried systems, (b)(f) examples of 3D cracks data with different vehicle vibrates up and down, (c)(g) examples of 3D cracks data with different pavement of different materials, (d)(h) examples of 3D cracks data with different cracks.

cracks in the original data, and an enhancement algorithm had to be generalized and applied to all heterogeneous 3D pavement data.

(2) The dynamic range and fluctuation of the obtained 3D data were quite different because of the random and unpredictable driving conditions as shown in Fig. 1. The cross slopes caused by the vehicle's vertical vibrations in a dynamic environment had to be uniformly removed to enhance the crack edges. Otherwise, the edge characteristics of the cracks are buried in the large-scale cross-slope profiles. Removing the influence of the dynamic environment and enhancing the crack edges are key to improving the generalization of crack-detection models.

(3) The differences caused by different pavement materials, textures, and defects caused the characteristics of cracks and backgrounds to show various shifts among the different data. Crack-detection methods should be adapted to 3D data with various pavement backgrounds.

In Fig. 2, the crack and non-crack characteristics (elevation and gradient maps) of typical 3D pavement data are illustrated at the profile and pixel levels. Many textures and real cracks exhibited shape characteristics in the elevation profile. It was difficult to distinguish cracks from the background textures using the elevation and gradient features shown in Fig. 2. Based on the original 3D data and typical features, the labeled samples cannot be used in other heterogeneous 3D data. These heterologous differences make it difficult to develop crack-detection methods based on traditional supervised learning that cannot overcome domain-shift problems between heterogeneous data (Azimi et al., 2020; Inoue and Nagayoshi, 2022). It is necessary to reduce the domain shifts caused by the acquisition parameters, pavement environments, backgrounds, and crack types to ensure that the limited labeled samples play a supervisory role in heterogeneous 3D pavement-crack detection in practical applications. Therefore, a 3D crack detection method with a TL ability is required.

2.3. Related works for domain shifts

TL and domain adaptation have attracted increasing attention in recent years owing to their ability to overcome the domain-shift problems between heterogeneous source and target domain data. Compared with the supervised method, the TL method has a more practical generalization ability for heterogeneous data. However, compared with the unsupervised method, TL can transfer existing knowledge to more tasks and improve efficiency.

Recently, a DNN with TL for 2D concrete-dam crack classification and a weakly supervised localization ImageNet were applied for pretraining (Li et al., 2022), indicating that the cross-domain TL crack-detection task was relatively challenging. A crack segmentation method for 2D masonry surfaces using a CNN and TL was proposed (Dais et al., 2021) and implemented by leveraging the effect of TL via fine-tuning. In Yang and Ji (2021), it was indicated that crack detection with TL ability is a practical requirement for engineering applications, and a two-stage framework using Unet++ and deep TL for 2D pavement crack detection was proposed. To use off-the-shelf datasets combined with preprocessing and pretraining to achieve cross-domain crack recognition is very challenging. Gopalakrishnan et al. (2017) used a DCNN pre-trained in ImageNet dataset to realize the crack detection of 2D asphalt/cement pavement through network fine-tuning. Similar studies have used ImageNet or related off-the-shelf datasets to extract 2D pavement cracks (Zhang et al., 2018b; Islam et al., 2022; Wu et al., 2021).

In summary, current crack detection studies have begun to focus on TL and generalization capabilities. Through preprocessing and fine-tuning, homogeneous and similar cross-domain transfer methods have been studied and applied to 2D infrastructure crack detection.

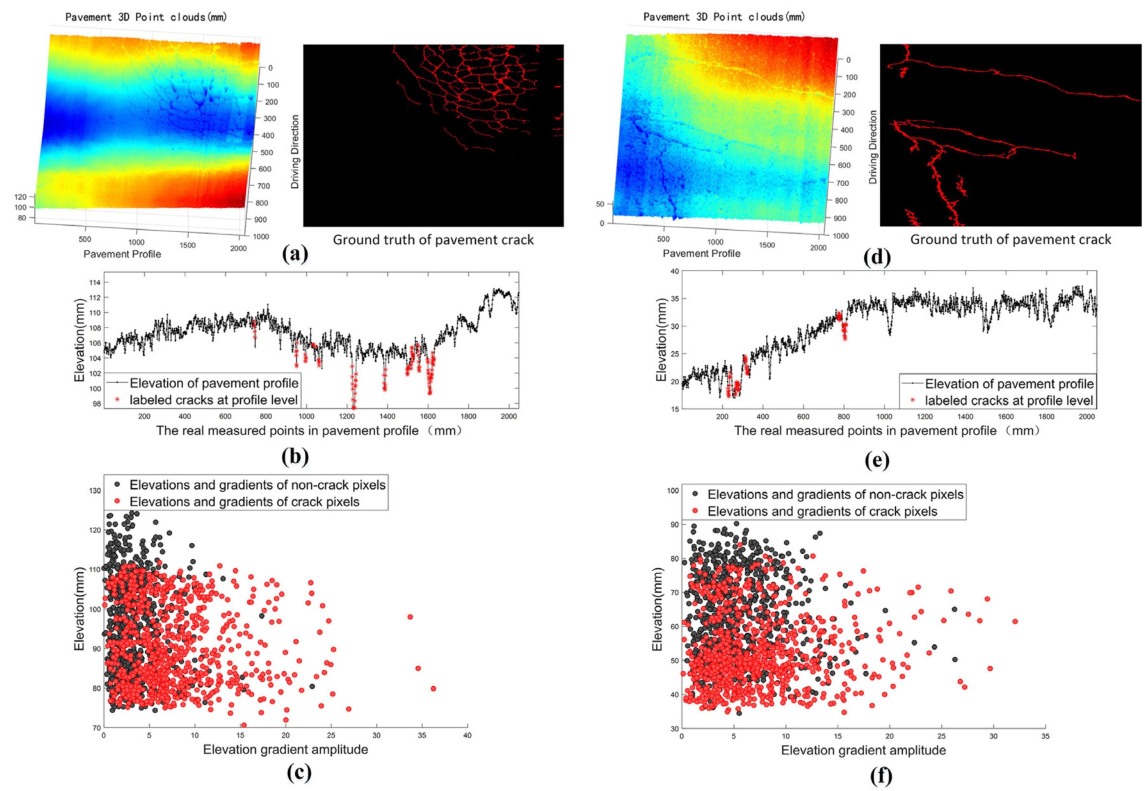

Fig. 2. The crack and non-crack characteristics of 3D pavement data and the domain shifts problems, (a)(d) are the 3D data and corresponding labeled ground truth, (b)(e)the crack and non-crack characteristics at 3D profile level, (c)(f) are the statistical characteristics of height/gradient of cracked pixels and non cracked pixels in original 3D data.

However, there has been little research on cross-domain 3D crack-detection tasks, and the current methods mainly achieve cross-domain detection through network fine-tuning. To some extent, fine tuning is not conducive to actual engineering applications, and current methods rarely involve domain-shift analysis of different cross-domain data.

3. Proposed cross-domain 3D pavement crack detection methodology

To overcome the domain shifts among the heterologous 3D data mentioned in Section 2, limited labeled samples play supervisory roles in more heterologous 3D data. The workflow of the proposed method is shown in Fig. 3. First, based on the profile frequency and sparse characteristics, the PCDM was applied to remove slowly varying and wide deformations caused by pavement deformation, and the vehicle vibrated vertically in the 3D laser scanning data. This step enhanced the crack characteristics at the laser scanning profile level. Second, crack-enhanced 3D data were transformed into 2D depth maps. Subsequently, based on a small number of labeled samples for training, the HED composed of the FCN and DSN can predict the probability of cracks in the depth maps of the heterogeneous 3D pavement data. Combined with the predicted probability and depth attribute information of the 3D data, the location and 3D attribute information of a crack can be automatically obtained. The multiscale and multilevel characteristics of depth edges caused by cracks in the 3D data were enhanced and used in the HED. Third, by utilizing the obtained probability and depth information in the 3D data, the crack location and 3D attributes can be obtained automatically. The proposed framework can reuse limited labeled samples in different pavement-crack detection tasks and, unlike unsupervised methods, does not require threshold parameters. The framework overcomes the domain shifts among different 3D pavements at two levels: the interference caused by the dynamic measurement environment and pavement deformations is reduced by the profile frequency and sparse characteristics, and the domain shifts caused by different pavement textures and crack types are overcome by the multiscale and multilevel feature learning of the HED network.

3.1. Improved 3D-PCDM

As illustrated in Fig. 1, several interference factors affected the crack characteristics in the original 3D data (size $M * N$). The most significant interferences were fluctuations caused by the vehicle's vertical vibrations in the driving environment, pavement deformation, and nonuniformity of pavement textures. Among them, the slow-varying wide fluctuations were macroscale in the profile, and the impact caused by the pavement texture was relatively microscale (Zhang et al., 2018c). To enhance and highlight the local sharp downward 'V' characteristics of cracks in the profile level, it was necessary to remove the macroscale fluctuations in the profile. The enhancement algorithm applied in the proposed method was based on the 3D PCDM in our previous work (Gui et al., 2018). First, the frequency and sparse characteristics of pavement distress and performance indicators were analyzed from the perspective of 3D pavement profiles $\mathbf{y}$. Second, a designed high-pass filter was employed to separate the low-frequency component $\mathbf{f}$ from the 3D pavement profile data. Third, the sparse component $\mathbf{x}$ and vibration component $\mathbf{t}$ were separated by total variation denoising based on the sparse characteristics. The low-frequency component $\mathbf{f}$ corresponds to pavement deformations or macroscale fluctuations, the sparse component $\mathbf{x}$ corresponds to the location of severe cracks, and the vibration component $\mathbf{t}$ corresponds to local fluctuations in the profile such as textures or thin and shallow cracks.

$$\mathbf{y}_i = [e_1, e_2, e_3, \dots, e_k], i \in [1, M], k \in [1, N] \tag{1}$$

where e_k is the elevation value of the profile point sequence,

$$\mathbf{y}_i = \mathbf{f}_i + \mathbf{x}_i + \mathbf{t}_i \tag{2}$$

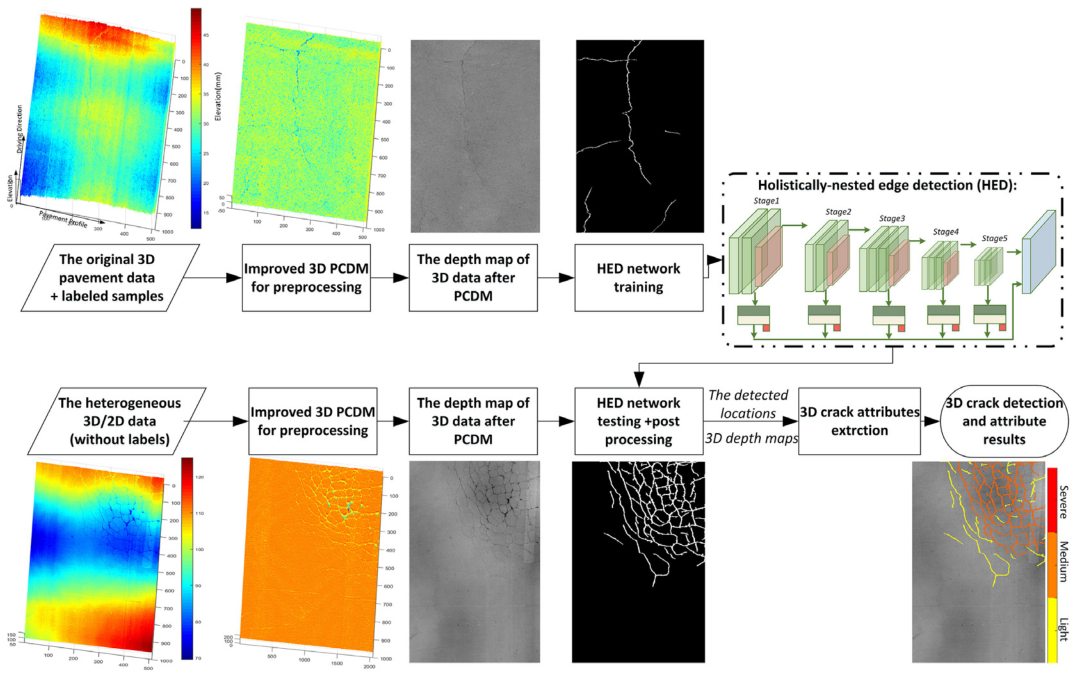

Fig. 3. Workflow of proposed PCDM-HED based on enhanced deep edge features.

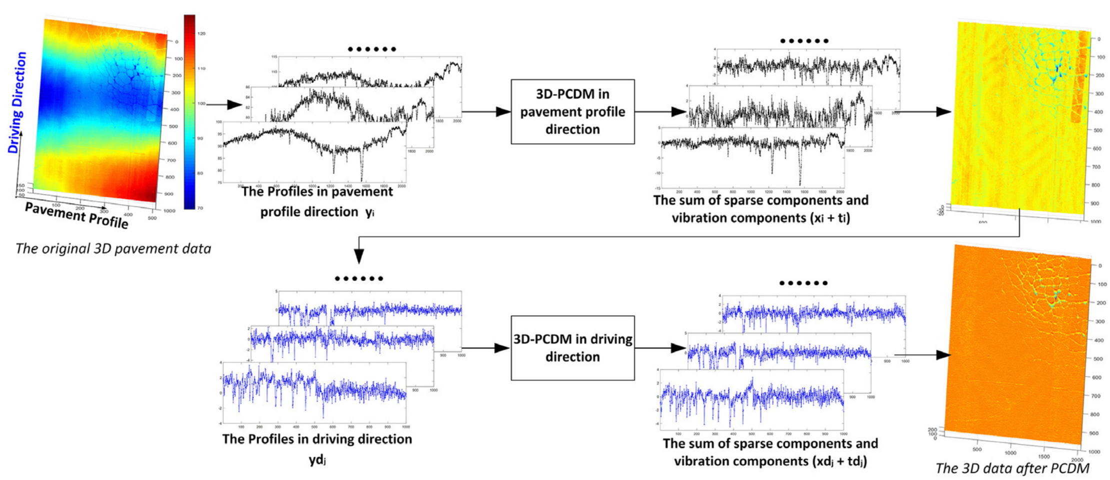

Fig. 4. The workflow and effect of improved 3D-PCDM.

The original 3D-PCDM in Gui et al. (2018) dealt only with pavement and macroscale fluctuations in the profile direction. According to the actual situation, there were also pavement deformations or macroscale fluctuations in the driving direction of the 3D pavement data. To completely remove this macroscale low-frequency interference and enhance the crack information in the 3D data, the original 3D-PCDM was improved. The workflow and effects of the improved 3D-PCDM are shown in Fig. 4.

The improved 3D-PCDM mainly includes the following two steps.

(1) The original 3D-PCDM was conducted along the pavement profile. To maintain the integrity of the crack structures and edges, high-frequency components (the sum of the sparse and vibration components) were retained. The high-frequency components of each profile were then spliced based on the following steps: These intermediate results included pavement deformations and macroscale fluctuations in the driving direction.

(2) The original 3D-PCDM was conducted in the driving direction. Subsequently, the high-frequency components of each profile were spliced in the driving direction. The obtained results of the improved 3D-PCDM not only removed the macroscale fluctuations caused by driving dynamics and pavement deformation defects but also enhanced the elevations and structures of the cracks without losing details. Fig. 5 illustrates the effect of improved 3D-PCDM processing from the perspective of 3D data and crack feature level.

Because the cut-off frequency of the designed filter comes from the generally applicable defect scale and data resolution, preprocessing can be applied to data with different fluctuation characteristics from different sources to significantly reduce the domain-shift problem caused by the fluctuation of heterogeneous data and enhance the crack features in the heterogeneous data.

As shown in Fig. 5, after preprocessing the above data using the improved 3D-PCDM, the crack features were enhanced and the domain shifts of the heterogeneous 3D pavement data were reduced to a certain extent; however, some distribution differences still existed. To ensure that the limited labeled samples play a role in more heterogeneous 3D data, it is necessary to further distinguish real cracks and the interference caused by pavement texture backgrounds.

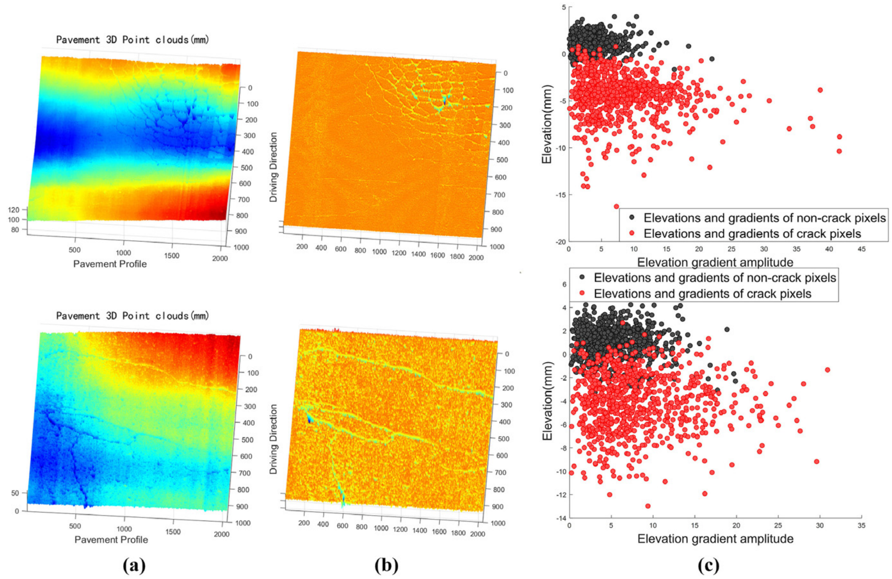

Fig. 5. The statistical characteristics of height/gradient of cracked pixels and non cracked pixels in the 3D data after PCDM processing.

3.2. HED network

After the improved 3D PCDM processing, the edge characteristics of the cracks were enhanced, and macroscale low-frequency interferences caused by driving dynamics and pavement deformation defects were removed. To segment cracks using a depth-learning network, a 3D depth map was proportionally mapped onto a 2D depth map; that is, the gray level of the transformed 2D image corresponded to the 3D depth map.

Pavement cracks have different thicknesses, depths, and types, and the crack characteristics of pavements with different structural depths and backgrounds are considerably different. It has been recognized that cracks in 2D pavement data exhibit multiscale and multilevel characteristics (Yan et al., 2022). Additionally, cracks exhibit sparse characteristics. Generally, the proportion of pavement cracks in the data is low. The selected method must consider the multiple types, sparse characteristics, different depths, and multiscale characteristics of cracks in 3D pavement data. In recent years, many crack-detection methods based on depth learning have achieved promising experimental results on specific datasets. Many existing crack-detection methods based on depth learning are sub-block classification methods. These methods can easily miss the local detection of weak cracks and do not consider the sparse and multiscale characteristics of cracks.

Cracks have edge characteristics in 2D and 3D data. The use of the deep learning method of edge detection to automatically detect cracks in pavement data is a development trend. HED (Heidler et al., 2022; Xie and Tu, 2015) is an end-to-end edge-detection method based on deep learning. Image-to-image prediction was performed using a deep learning model with a complete convolution neural network and a deep supervision network. HED automatically learns the types of rich hierarchical features under the guidance of deep supervision of the side response, which is close to the human ability to solve the edge fuzziness of natural images and target boundary detection. By combining multiscale and multilevel visual responses, HED shows good results in image-to-image learning, emphasizing the inherited and gradually refined edge map generated as a side output. An HED network design takes into account the fact that edge pixels and non-edge pixels are seriously unbalanced, and uses a simple strategy to automatically balance the loss between positive/negative classes. This class balance weight is used as a simple method to offset the imbalance between edges and non-edges. The network structure diagram of HED in Fig. 6. shows that this network fully considers the multiscale, multilevel, and sparse characteristics of pavement cracks.

Suppose the HED network has P ($P = 5$ in Fig. 6) side-output layers, the corresponding weights are denoted as $\mathbf{w} = (\mathbf{w}^{(1)}, \ldots, \mathbf{w}^{(p)})$, and the objective function is following,

$$L_{side}(\mathbf{W}, \mathbf{w}) = \sum_{p=1}^{P} \alpha_p l_{side}^{(p)}(\mathbf{W}, \mathbf{w}^{(p)}) \tag{3}$$

where $l_{side}^{(p)}$ denotes the image-level loss function for side-outputs. In HED training, the loss function is computed over all pixels in a training depth map X and edge map Y. During testing, given the depth map, the edge map predictions were obtained from both the side output layers and the weighted fusion layer as

$$(\hat{Y}_{fuse}, \hat{Y}_{side}^{(1)}, \ldots, \hat{Y}_{side}^{(P)}) = CNN(X, (\mathbf{W}, \mathbf{w}, \mathbf{h})^{*}) \tag{4}$$

where $CNN()$ denotes the edge map produced by the HED network. The final unified output was obtained by further aggregating the generated edge maps as

$$\hat{Y}_{HED} = Average(\hat{Y}_{fuse}, \hat{Y}_{side}^{(1)}, \ldots, \hat{Y}_{side}^{(P)}) \tag{5}$$

Based on the above 3D data for removing the crack with vehicle attitude fluctuations, the crack presents the characteristics of the depth edge in the 3D depth data. The 3D depth data were transformed into a depth map, normalized, and input into the HED network. When training the HED, the crack labeling and input data were at the map level. Testing the HED resulted in five levels of edge outputs and one fusion result. Among the six outputs, cracks with different thicknesses were observed. These intermediate results formed the basis for subsequent crack identification and information extraction. The output position and effect of each layer feature in the HED network are shown in Fig. 6.

3.3. Post processing and 3D crack information extraction

The HED network obtains the fracture boundary probability and not the determined fracture/non-fracture binary map. To obtain the 3D attribute fracture information (depth and distribution), the probability map must be transformed into a fracture binary map. Here, CFAR detection was used to automatically obtain information regarding the crack location. Based on the location information combined with the high-frequency component depth map obtained by the 3D PCDM, the 3D depth information and distribution information of the fractures can be obtained. According to the depth threshold, the severity of 3D pavement cracks is evaluated automatically.

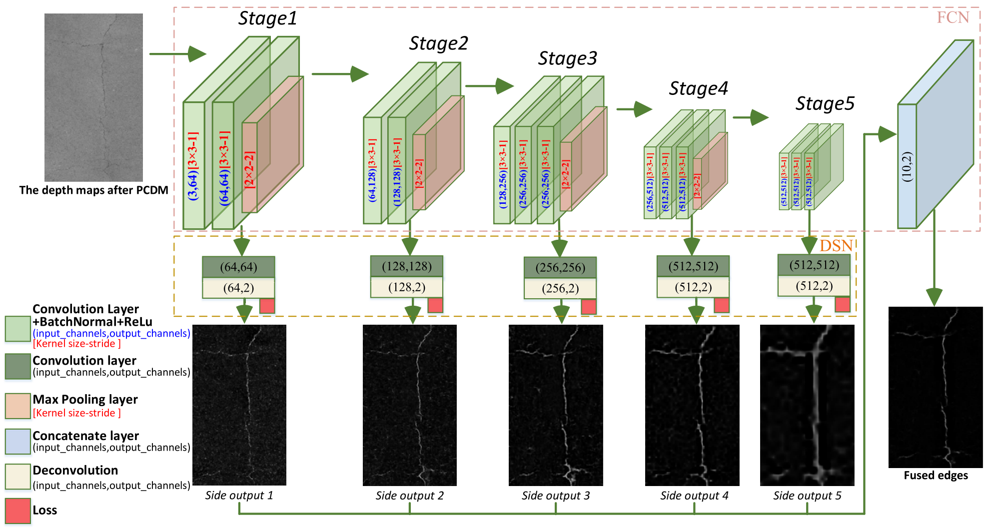

Fig. 6. The network structure diagram of HED and the effect of each layer.

Table 1
The basic information of applied datasets.

Dataset name and usages	Acquisition system	Crack and background characters	Labeled ground truth	Sample number and size
LS-3D-1, **training**	Laser-scaning 3D	Asphalt pavement, thin cracks	All labeled	120(1000*512)
LS-3D-2, **testing**	Laser-scaning 3D	Asphalt pavement, thin cracks	All labeled	30(1000*512)
LS-3D-3, **testing**	Laser-scaning 3D	Asphalt pavement, alligator cracks	Unlabeled	212(1000*512)
LS-3D-4, **testing**	Laser-scaning 3D	Concrete pavement, block cracks	Unlabeled	208(1000*512)
LCMS-I, **testing**	Laser-scaning	Asphalt pavement, block cracks	Partial labeled	65(700*800)
ESAR, **testing**	2D image	Asphalt pavement, uncontrolled lighting	Partial labeled	30(512*768)
2D-I, **testing**	2D image	Asphalt and, concrete pavement	All labeled	250(384*544)

CFAR detection is a rapid, efficient, and stable method for target detection in SAR images (El-Darymli et al., 2013). It has been proved (Gui et al., 2019) suitable for the rapid determination of 3D pavement cracks. The 3D PCDM introduced in Section 3.1 eliminates the interference of driving fluctuations and other noises in the original 3D data, and the depth and distribution of cracks are automatically obtained by integrating the location of cracks and the 3D pavement depth information. Here, we divided the crack depth into light, medium, and heavy levels according to the depth threshold standard in Gui et al. (2019) and Wen et al. (2022b). Information on crack distribution and severity is highly significant for quickly and comprehensively grasping the degree of damage and safety performance evaluation of pavement and for maintenance decision-making.

4. Experimental results and analysis

This section introduces the datasets used and their settings. The experimental results of crack detection are presented, analyzed, and compared for quantitative evaluation, and extended experimental results of 3D crack attribute analysis are presented. Finally, the parameter sensitivity and limitations of the proposed method are discussed.

4.1. Datasets descriptions and settings

To validate the effectiveness and practicability of the proposed TL 3D crack detection method, sufficient crack datasets from 3D laser scanning systems (including LSP and LCMS) were collected. Because the proposed method combines the generalization of TL and effectiveness of deep learning, two classic 2D datasets were used to verify the applicability and scalability of the proposed method for 2D pavement crack detection.

The basic information and typical data examples of applied datasets are shown in Table 1, and the typical examples of each dataset are shown in Figs. 7–13. Specifically, the laser-scanning 3D pavement data (The sample data is shown in Figs. 7–10) were collected by LSP equipments, the 3D crack are distributed in several real road sections in China. These 3D datasets include cracks with different pavement damage degrees and types. For the applied 3D datasets, the resolution in the elevation direction was about 0.25 mm, and the resolution in the profile direction was about 1 mm, and that in the driving direction varied from 1 to 5 mm. For the laser scanning intensity dataset LCMS-I, the reflected intensity, not elevation can also reflect the location of the crack. As shown in Fig. 11, although the laser scanning intensity data is not affected by the light shadow, there are uneven laser distributions. For the applied 2D pavement datasets, the uneven lighting and shadows in the actual road environments are also included in the datasets. This kind of uneven light also has some influence on the crack extraction, and the proposed method tries to use PCDM model to remove this factor.

The basic information and typical data examples of the applied datasets are listed in Table 1, and typical examples of each dataset

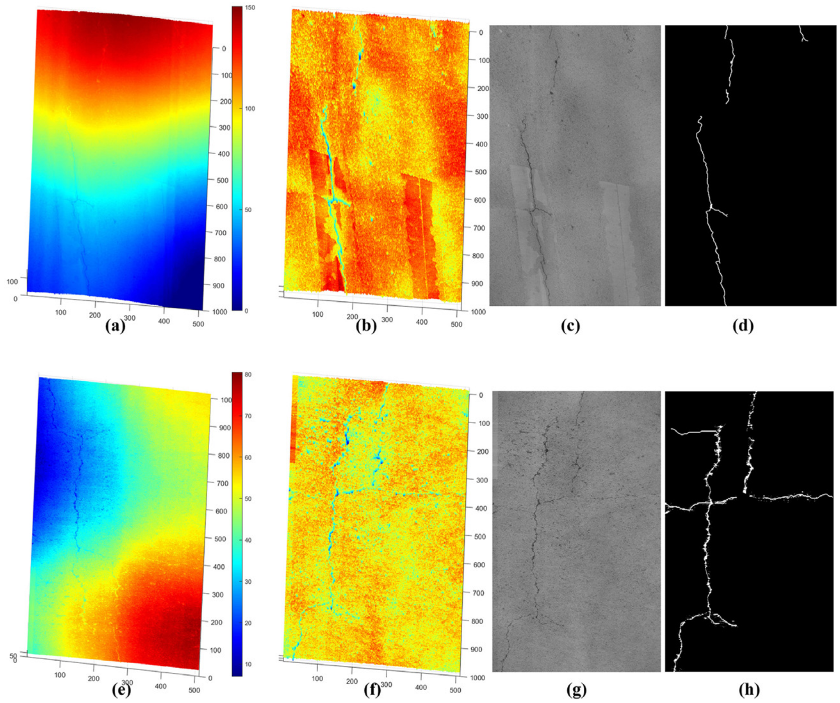

Fig. 7. The typical data examples from LS-3D-1, (a)(e) are the original 3D data, (b)(f) are the 3D data after PCDM processing, (c)(g) are the depth map of (b)(f), (d)(h) are the labeled crack ground truth.

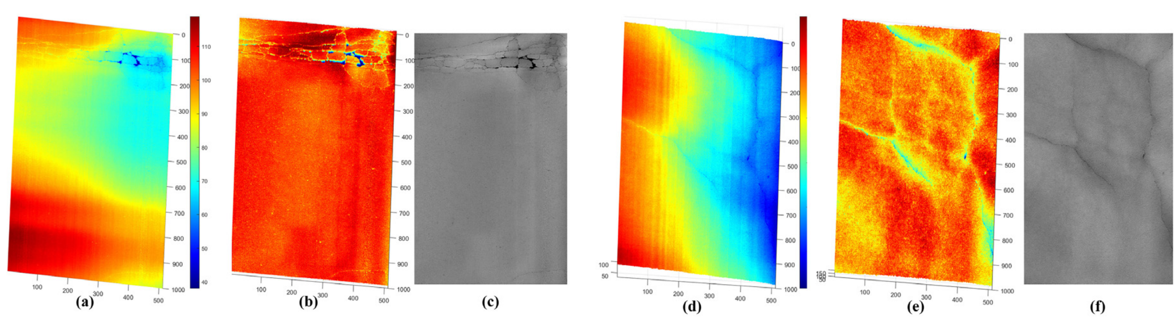

Fig. 8. The typical data examples from LS-3D-2, (a)(d) are the original 3D data, (b)(e) are the 3D data after PCDM processing, (c)(f) are the depth map of (b)(e).

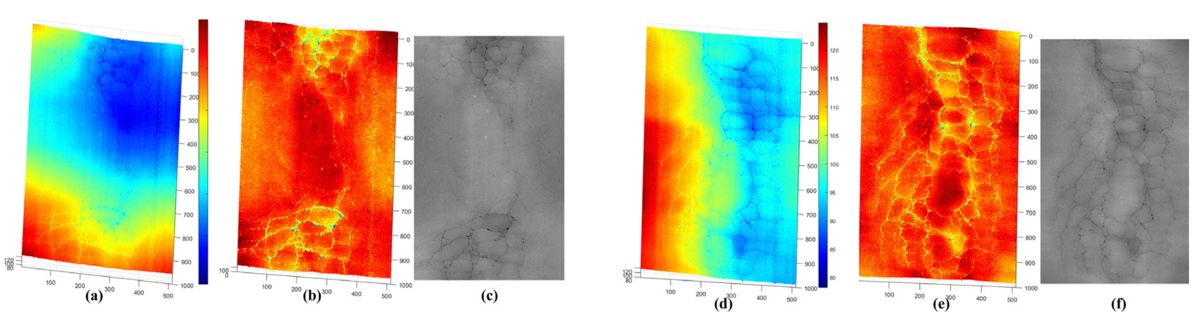

Fig. 9. The typical data examples from LS-3D-3, (a)(d) are the original 3D data, (b)(e) are the 3D data after PCDM processing, (c)(f) are the depth map of (b)(e).

are shown in Figs. 7–13. Specifically, the sample laser-scanned 3D pavement data shown in Figs. 7–10 were collected by an LSP system, and the 3D cracks were distributed in different real road sections in China. These 3D datasets included cracks of different degrees and types of pavement damage. For the applied 3D datasets, the resolution in the elevation direction was about 0.25 mm, and the resolution in the profile direction was about 1 mm, and that in the driving direction varied from 1 to 5 mm. For the laser scanning intensity dataset LCMS-I, the reflected intensity, and not the elevation, reflected the cracks. As shown in Fig. 11, although the laser scanning intensity data were not affected by a light shadow, an uneven laser distribution was observed. For the applied 2D pavement datasets, uneven lighting and shadows in actual pavement environments were included. This uneven light has a negative influence on crack extraction, and the proposed method attempts to use the PCDM to remove this factor.

The above experimental data included sufficient crack types as well as different 3D and 2D pavements (sample data are shown in Figs. 11–13). As shown in Figs. 7–13, the cracks in the 3D and 2D

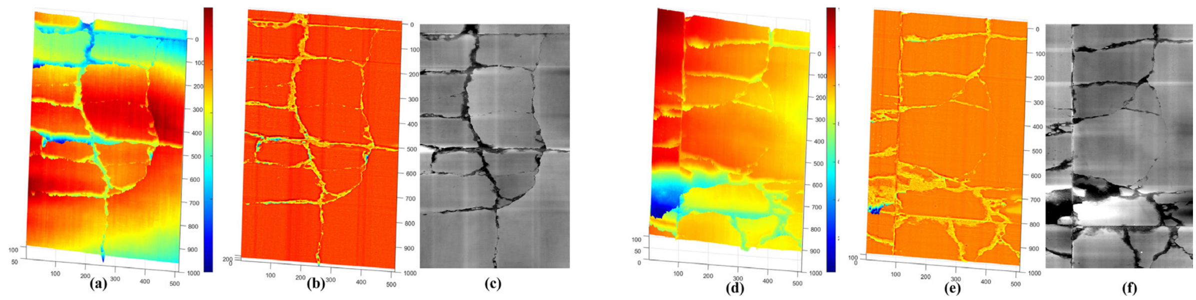

Fig. 10. The typical data examples from LS-3D-4, (a)(d) are the original 3D data, (b)(e) are the 3D data after PCDM processing, (c)(f) are the depth map of (b)(e).

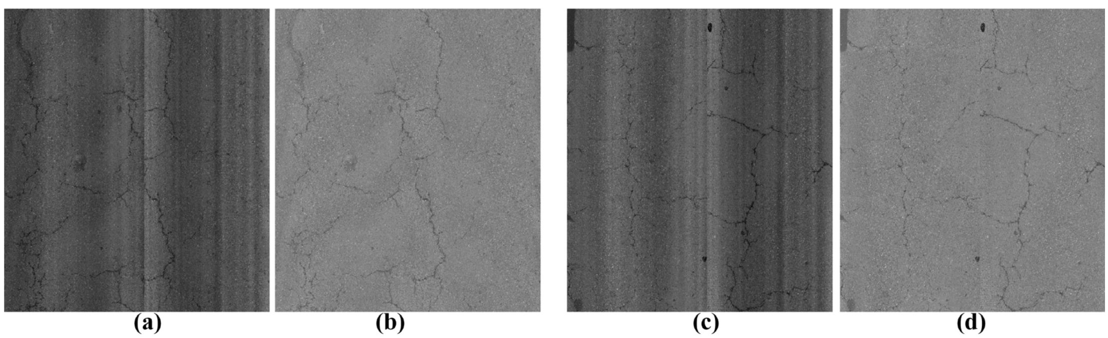

Fig. 11. The typical data examples from LCMS-I, (a)(c) are the original data, (b)(d) are the data after PCDM processing.

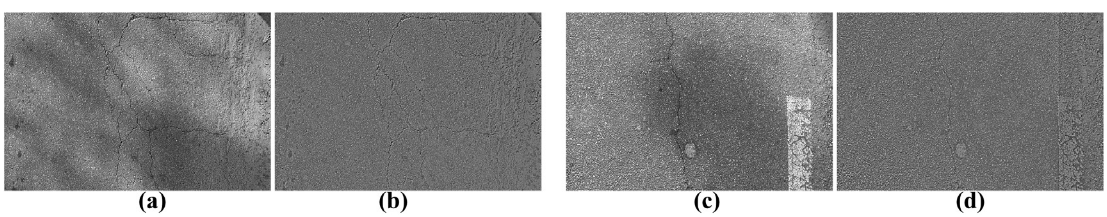

Fig. 12. The typical data examples from ESAR, (a)(c) are the original data, (b)(d) are the data after PCDM processing.

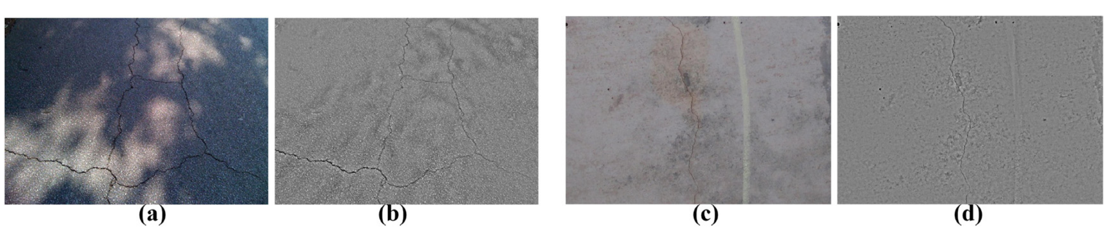

Fig. 13. The typical data examples from 2D-I, (a)(c) are the original data, (b)(d) are the data after PCDM processing.

data after PCDM processing were more obvious than those in the original data. The edge features of the cracks in the original 3D data were not obvious because of the vehicle fluctuations and pavement deformation. The characteristics of the cracks in the original 2D data were also not obvious because of the uneven lighting and shadows. The applied PCDM can remove these low-frequency impacts and highlight the characteristics of cracks in 3D or 2D pavement data. We used only the 3D dataset LS-3D-1 for training and combined it with the proposed method to test the other datasets. The results and analysis follow.

4.2. Quantitative performance for 3D and 2D crack extraction

Illustrations and crack detection results are shown in Figs. 14 and 15 and 19–23. Because the applied experimental data were mainly obtained from practical pavement maintenance engineering, accurate ground truths (GTs) were difficult to obtain, and it is unrealistic to obtain all the GTs for all the datasets. Thus, only four testing datasets (LS-3D-2, 2D-I, LCMS-I, and ESAR) were subjected to pixel-level quantitative evaluation to prove the effectiveness of the proposed method. By comparing the detected cracks with the GTs, the effective buffered Hausdorff (BH) metric (Gui et al., 2018, 2019; Jiang and Tsai, 2016), F-score, and recall score (Fei et al., 2019; Tang et al., 2021) were applied as quantitative evaluation indicators for the crack extraction results. The buffered Hausdorff metric scores ranged from 0 to 100, where 100 indicates perfect crack detection and location. For the experimental LS-3D-2 datasets, the accuracy of crack detection is shown in Figs. 16–18 and Table 2.

To further validate the effectiveness of the proposed PCDM-HED, the following comparative methods and experiments were conducted.

(1) Modified Cracktree: The acknowledged Cracktree method (Zou et al., 2012) (available online: http://cracktree.net/) was originally

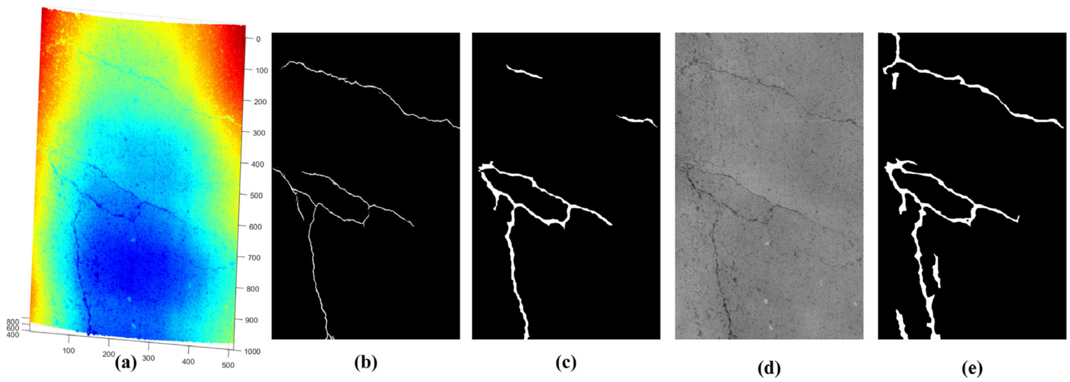

Fig. 14. The results and comparisons of samples from LS-3D-2, (a) the ordinal 3D data, (b) the results of Modified Cracktree, (c)the results of HED without PCDM processing, (d) the depth map after PCDM processing, (e)the results of proposed PCDM-HED.

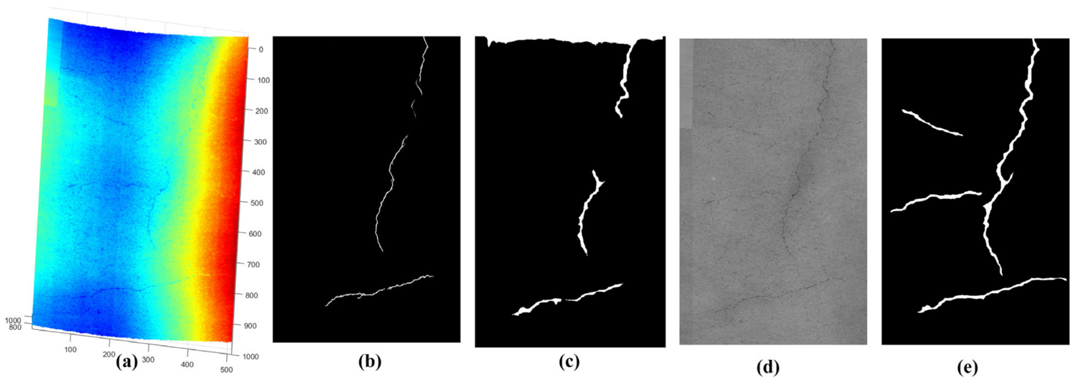

Fig. 15. The results and comparisons of samples from LS-3D-2, (a) the ordinal 3D data, (b) the results of Modified Cracktree, (c)the results of HED without PCDM processing, (d) the depth map after PCDM processing, (e)the results of proposed PCDM-HED.

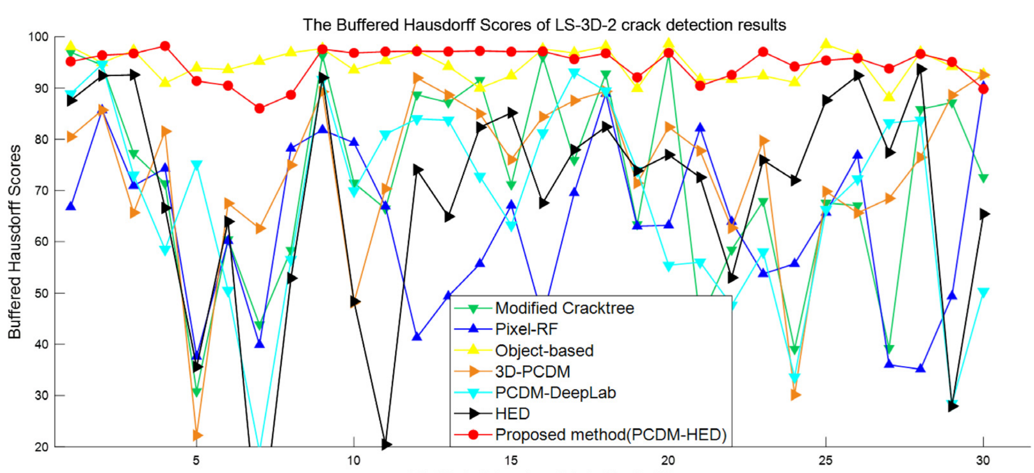

Fig. 16. The buffered Hausdorff scores of test dataset LS-3D-2 results and comparisons.

proposed to process 2D crack data. Here, the input of Cracktree was the converted 2D depth data based on data after PCDM.

(2) Pixel-RF: The random forest (RF)-based method was used to compare the proposed method with the traditional machine learning approach (Zhang et al., 2017b). In this pixel-level classification based on RF, a sliding window of 15*15 pixels was centered at each pixel, and the local context around each pixel was continuously scanned. The same local features as those applied in Zhang et al. (2017b), including the minimum value, mean, variance, and third-, fourth-, and fifth-order moments, were extracted for each pixel. As this traditional method cannot transfer or self-adaptively classify data with different textures and depths, 20% of the crack pixels and 10% of the background pixels were randomly selected from each manually labeled GT as training samples. These labeled pixels were used to train the RF, and the pixel-RF completed the pixel-level crack extraction. The input of Pixel-RF was the converted 2D depth data based on the data obtained after PCDM.

(3) Object-based (Gui et al., 2019): The object-based image analysis method for crack detection and attribute extraction from laser-scanned 3D profile data is primarily based on the local statistical characteristic differences between crack objects and surrounding texture objects. This

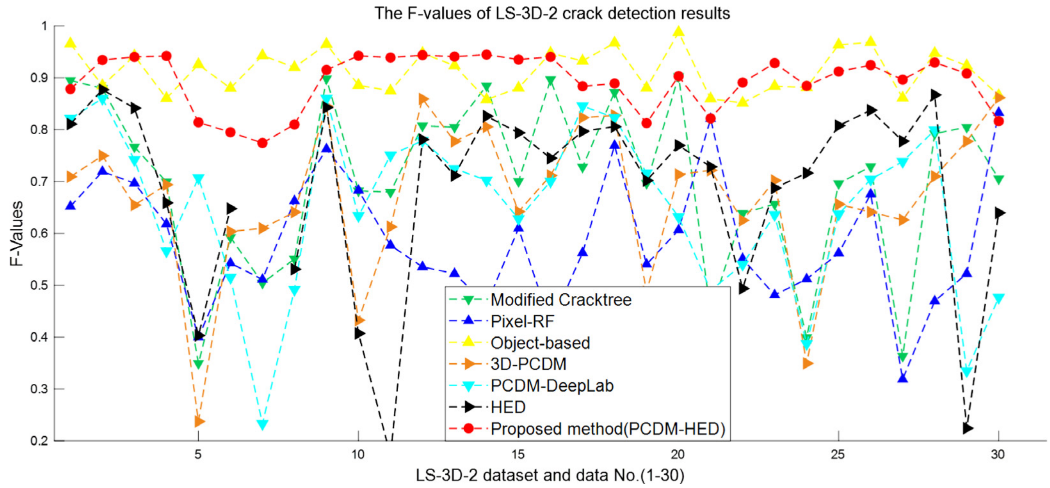

Fig. 17. The F-values of test dataset LS-3D-2 results and comparisons.

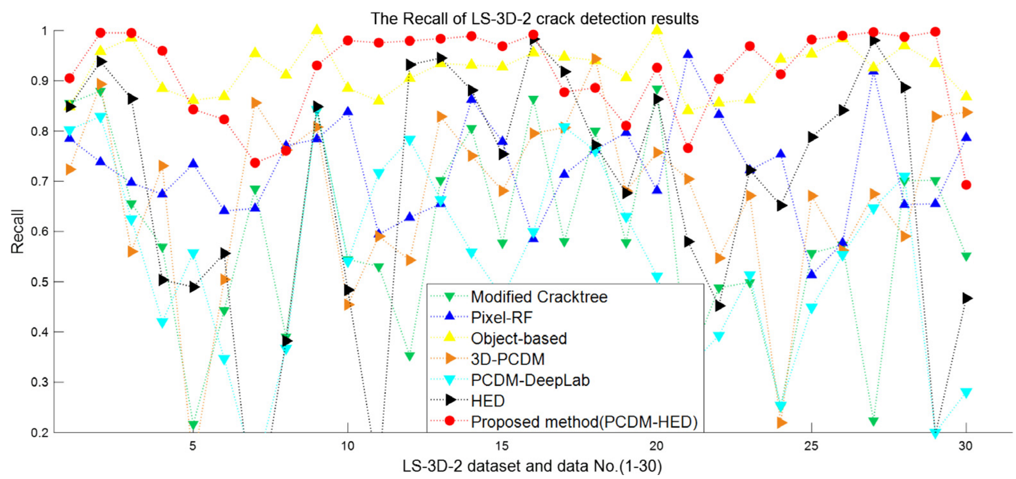

Fig. 18. The recall of test dataset LS-3D-2 results and comparisons.

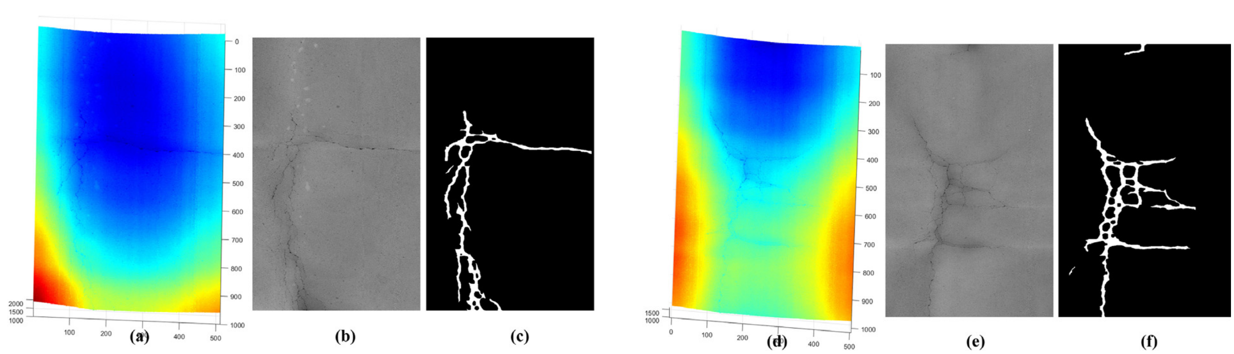

Fig. 19. The results and comparisons of samples from LS-3D-3, (a)(d) the ordinal 3D data, (b)(e) the depth map after PCDM processing, (c)(f) the results of proposed PCDM-HED.

method has a low degree of automation and requires considerable threshold determination and is not conducive for large-scale automatic crack detection.

(4) 3D PCDM (Azimi et al., 2020): Cracks are detected by analyzing sparse components from pavement profile data. In the crack judgment step, the automation degree of this method was low.

(5) Unet (Ronneberger et al., 2015): This classic segmentation network has certain advantages for edge and crack extraction. Here, it was used to compare the model with one of the benchmark methods to highlight the effectiveness of the applied HED.

(6) PCDM-Unet: With this combination of PCDM preprocessing with Unet, it was necessary to further highlight the effects of preprocessing and the selected HED because of the significant vehicle fluctuations and pavement deformation in the original 3D data.

(7) DeepLab (Chen et al., 2017): DeepLab is a common segmentation network used as a benchmark.

(8) PCDM-DeepLab: PCDM preprocessing combined with DeepLab.

(9) HED (Heidler et al., 2022; Xie and Tu, 2015): HED trained without PCDM preprocessing.

Among these comparison methods, the modified Cracktree, object-based, and 3D PCDM can be classified as unsupervised methods, the

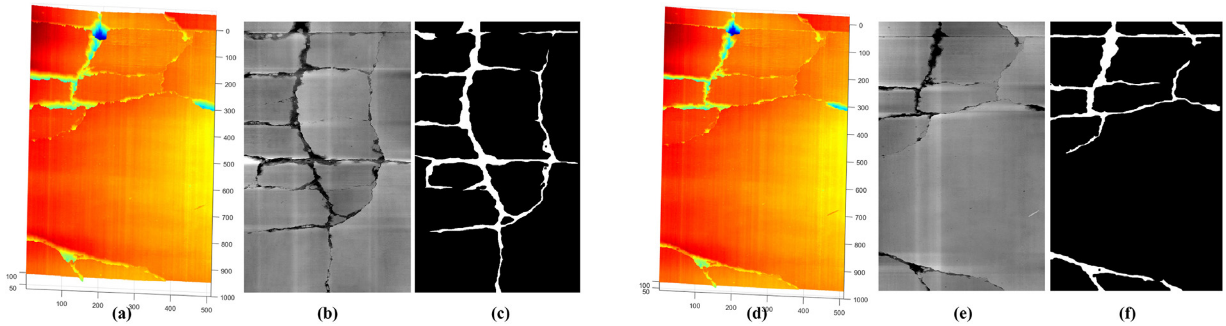

Fig. 20. The results and comparisons of samples from LS-3D-4, (a)(d) the ordinal 3D data, (b)(e) the depth map after PCDM processing, (c)(f) the results of proposed PCDM-HED.

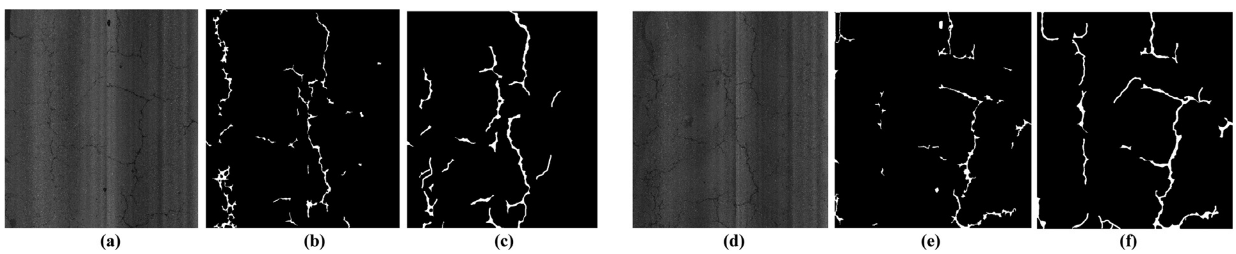

Fig. 21. The results and comparisons of samples from LCMS-I, (a)(d) the ordinal data, (b)(e) the results of HED without PCDM processing, (c)(f) the results of proposed PCDM-HED.

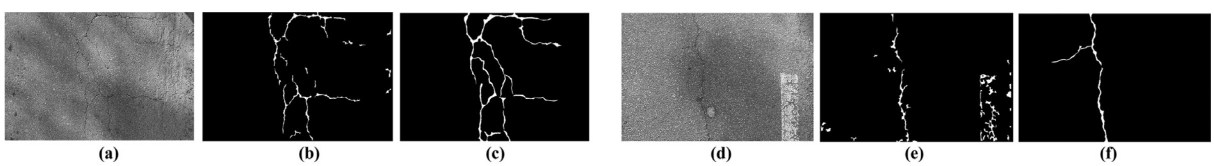

Fig. 22. The results and comparisons of samples from ESAR, (a)(d) the ordinal data, (b)(e) the results of HED without PCDM processing, (c)(f) the results of proposed PCDM-HED.

Table 2
Accuracy evaluations of LS-3D-2 datasets by different methods (the mean values of evaluation indicators).

Method	**Modified Crack**	**Pixel-RF**	**Object -based**	**3D-PCDM**		
mean BHS	72.0154	63.1309	94.5315	73.8976		
mean F-values	0.7007	0.5867	0.9025	0.6702		
mean Recall	0.5877	0.7243	0.9100	0.6704		
Method	**Unet**	**PCDM-Unet**	**DeepLab**	**PCDM-DeepLab**	**HED**	**Proposed PCDM-HED**
mean BHS	15.3691	38.2882	25.8494	67.8298	68.5241	**94.7432**
mean F-values	0.1721	0.3836	0.2698	0.6491	0.6634	**0.8929**
mean Recall	0.4217	0.2852	0.3852	0.5427	0.6771	**0.9171**

Pixel-RF is a data-supervised method, and the training test data of these methods were homogeneous. Moreover, the experimental setups of HED, Unet, PCDM-Unet, DeepLab, and PCDM-DeepLab were similar to the experimental setup of the proposed PCDM-HED, which belongs to the TL method on the LS-3D-1 dataset and testing on the LS-3D-2 and 2D-I datasets. The classic U-Net and DeepLab methods were selected for comparison to demonstrate the effectiveness of the applied PCDM and HED methods. PCDM-Unet, PCDM-DeepLab, and PCDM-HED were set up to fully highlight the effectiveness of PCDM and HED.

The results of the LS-3D-2 dataset and the corresponding quantitative evaluation and comparisons in Table 2 and Figs. 16–18 demonstrate the effectiveness of the proposed PCDM-HED. Due to the extremely uneven proportion of crack and non-crack regions in the 3D data, typical segmentation networks such as Unet and DeepLab are not suitable for this task. However, after adding PCDM preprocessing, the detection effect of PCDM-DeepLab has significantly improved. For the proposed PCDM-HED and the HED, Unet, PCDM-Unet, DeepLab, and PCDM-DeepLab methods, the labeled samples of the LS-3D-2 dataset did not participate in the training and were only used to verify the accuracy of the transfer method.

Furthermore, for datasets LS-3D-3, LS-3D-4, LCMS-I, ESAR, and 2D-I, the partial results are shown in Figs. 19–23. And the quantitative evaluations of the proposed PCDM-HED on these datasets are shown in Table 3. The results of these datasets further prove the transferability and generalization of the PCDM-HED. The results were evaluated and compared with the 2D-I dataset as shown in Fig. 24. The labeled annotations of the above test datasets were only used to verify the effectiveness and accuracy of the proposed TL method, and these labeled samples did not participate in the model training. In other words, the trained model can be used directly to test the dataset without any labeled samples, which is consistent with the actual requirements.

4.3. Extended experiments analysis

For the 3D pavement data, the proposed method can obtain the attribute information of cracks according to the detection and depth information as illustrated in Fig. 25. Because the crack locations were

Fig. 23. The results and comparisons of samples from 2D-I, (a)(d) the ordinal data, (b)(e) the results of HED without PCDM processing, (c)(f) the results of proposed PCDM-HED.

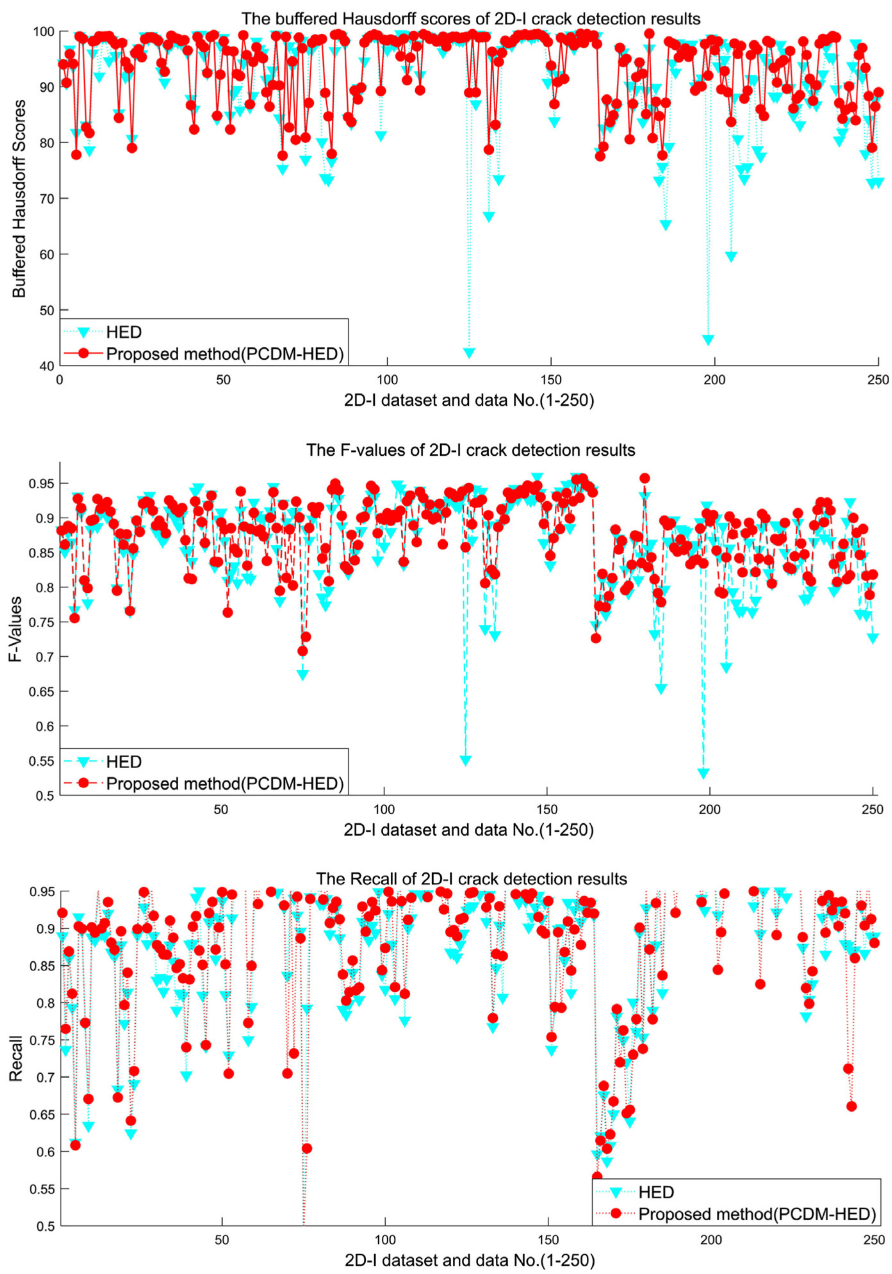

Fig. 24. The buffered Hausdorff scores, F-values and recall of test dataset 2D-I results.

detected based on the 3D depth and detected edges, this study used independent cracks as the basic unit for the attribute statistics. Because it is difficult to obtain true measurements of actual crack depth, the depth accuracy of the LSP measurement system was demonstrated in our previous study (Zhang et al., 2018c; Gui et al., 2019). Subsequently, based on the obtained 3D information, the severity of the cracks was comprehensively evaluated using area and depth information. Repeated crack attributes can be beneficial for calculating the health status and degree of defects in pavement. This severity information can be used as a basis for decision-making in pavement maintenance departments.

4.4. Discussion on method sensitivity and limitations

The main steps of the PCDM-HED include preprocessing, network prediction, and post-processing. Because of the method's transferability, it is generally possible to use the same set of parameters for

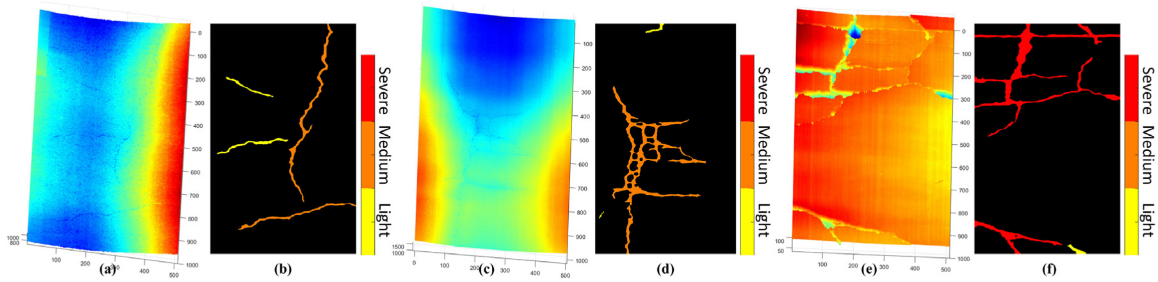

Fig. 25. Example of 3D crack attribute acquisition results, (a)(c)(e) the original 3D data, (b)(d)(f)the crack attributes.

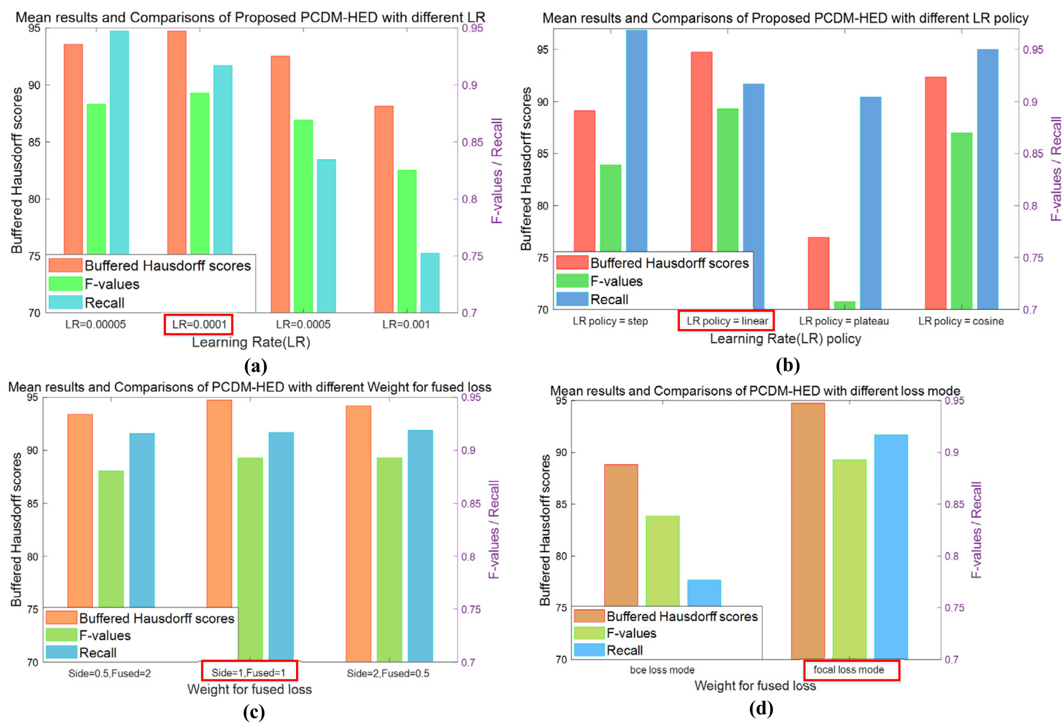

Fig. 26. Sensitivity analysis of different network hyperparameters based on LS-3D-2 dataset transfer test, (a) Learning rate adjustment comparisons, (b) Learning rate policy adjustment comparisons, (c) Weight for fused loss adjustment comparisons, (d) Loss mode adjustment comparisons.

Table 3
Accuracy evaluations of different datasets (the mean values of evaluation indicators).

Dataset	LS-3D-2	LCMS-I	ESAR	2D-I
mean BHS	94.7432	90.1725	96.4292	93.6994
mean F-values	0.8929	0.8463	0.8708	0.8773
mean Recall	0.9171	0.8419	0.8716	0.8919

different datasets, and it is not advisable to adjust the parameters. Specifically, both preprocessing and post-processing are unified, and the frequency characteristic parameter settings for preprocessing have been extensively verified in Gui et al. (2018) and exhibit good stability. The PCDM-HED network helps obtain accurate crack probabilities, and the CFAR detector used in post-processing is also relatively robust. The enhanced deep edge feature is the core of the effectiveness and generalization of the PCDM-HED; therefore, the key parameters of the method are network hyperparameters.

To further demonstrate the sensitivity and limitations of the PCDM-HED, we analyzed some network hyperparameters, including the adjustment of the network learning rate, learning rate rules, weight of fused loss, and loss mode, as shown in Fig. 26. The optimized network default hyperparameter settings are as follows: learning rate is set as 0.0001, learning rate policy is set as linear, weight of fused loss is set as 1/1, and loss mode is set as focal. To demonstrate the overfitting of the proposed method, we compared the loss distributions of training under different settings and performed dataset expansion verification and the corresponding loss analysis, as shown in Fig. 27. The comparison includes HED network training loss map without PCDM pre-processing, PCDM-HED network training loss map based on the original training set, PCDM-HED network training loss map based on the flipped expanded training set. Experiments show that PCDM pre-processing is conducive to avoiding overfitting, and increasing samples is also conducive to network convergence.

The stability of the method proposed in this paper has been demonstrated through comparative experiments adjusting the above parameters. However, the robustness of the current method is based on enhanced deep edge features. Due to the incomplete of labeled datasets, the edges are not entirely caused by cracks. Therefore, this method also has certain limitations. Here are specific examples to illustrate these limitations.

(1) Some no-crack objects with edges will be detected by PCDM-HED, such as joints in cement pavements, grooves in airport pavements, manhole covers and speed bumps in pavements. As shown in Figs. 28

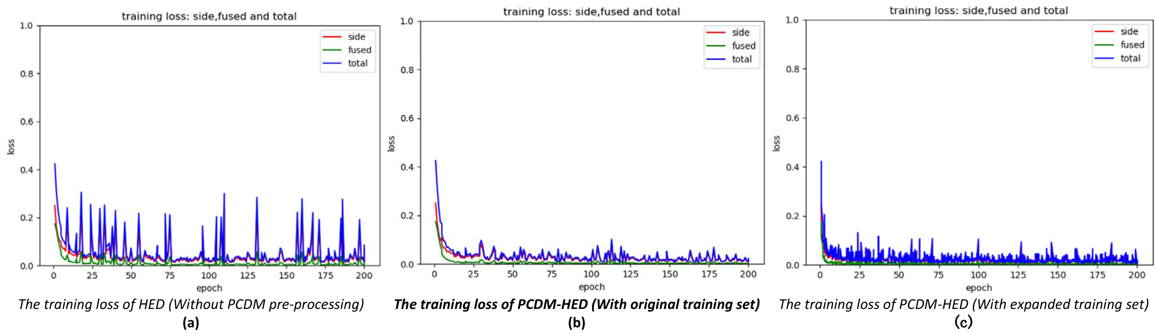

Fig. 27. Sensitivity analysis and comparisons of network training overfitting: (a) HED network training loss map without PCDM pre-processing, (b) PCDM-HED network training loss map based on the original training set, (c) PCDM-HED network training loss map based on the expanded training set.

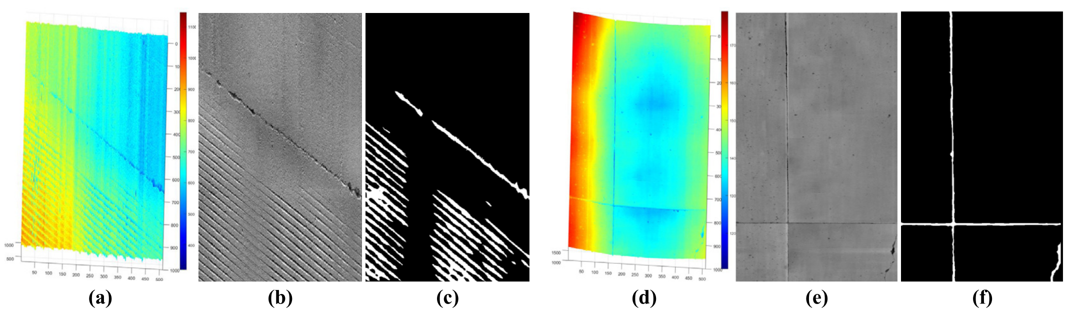

Fig. 28. Example of failure cases, (a) an airport groove 3D data with cracks, (b) the PCDM pre-processing depth map of (a), (c) the detection results. (d) a cement 3D pavement data with joints and cracks, (e) the PCDM pre-processing depth map of (d), and (c) is the detection result.

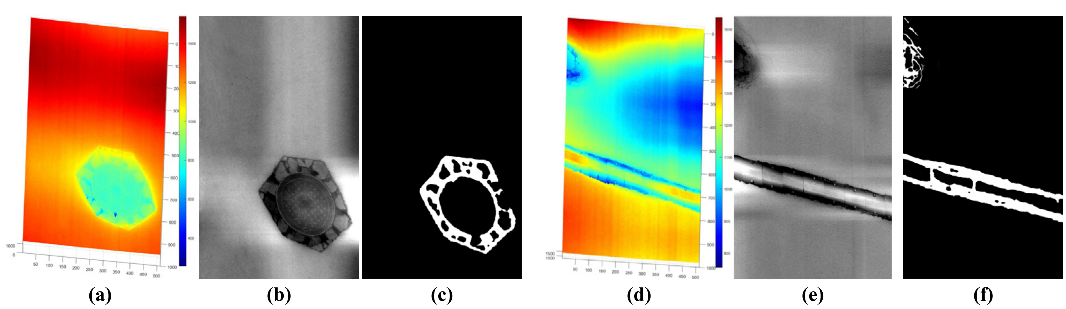

Fig. 29. Example of failure cases, (a) a 3D asphalt pavement data with manhole covers, (b) the PCDM pre-processing depth map of (a), (c) the detection results. (d) a 3D asphalt pavement data with speed bumps and cracks, (e) the PCDM pre-processing depth map of (d), and (c) is the detection result.

and 29. These failed cases are mainly caused by incomplete training datasets. These misidentifications are detrimental to the decision-making of pavement maintenance. Although the probability of these false detections is not high and is related to specific scenarios, this problem can be solved if the training dataset is improved by corresponding annotated samples.

(2) Based on the results described in Sections 4.2 and 4.3, the detection results and severity attributes of cracks are currently separate, due to the fact that the labeled dataset only annotates the location information. If subsequent annotations have corresponding crack severity information, combined with crack location and depth information, it is possible to automatically obtain both crack locations and severity attributes simultaneously.

(3) Due to the significant domain shifts in heterogeneous 3D data, the HED-PCDM adopts preprocessing and network multi-layer edge enhancements to overcome the domain shifts. Although the stability of the parameters is fully verified, to some extent, the automation level of this combination is not as good as that of end-to-end networks. In the future, it can be considered to integrate PCDM into the deep network, to further adjust and improve the network, and achieve an integrated heterogeneous data crack identification network.

5. Conclusion

Surging homogeneous 3D pavement data poses significant challenges for large-scale deep-learning-based crack-detection tasks. On the one hand, it is difficult to obtain sufficient labeled samples that fully conform to the test data; on the other hand, it is difficult to transfer accumulated labels to newly obtained data. Therefore, current crack-detection tasks require high robustness and transfer efficiency. In this context, this study proposed a robust and generalized framework for cross-scene 3D pavement-crack detection and attribute extraction called the PCDM-HED. The key to the PCDM-HED is the enhanced deep edge features, and the PCDM preprocessing enhanced the crack characteristics by the profile frequency and sparse characteristics of cracks. The HED network predicted the probability of cracks using multiscale and multilevel edge characteristics. The proposed framework implemented limited off-the-shelf labeled samples for use in different pavement-crack detection tasks and ensured robustness and generalization. Sufficient experimental verification and comparisons were carried out for cross-domain pavement-crack-detection tasks. Based on a 3D dataset with 120 labeled samples, the proposed PCDM-HED provided TL and identified test-unlabeled datasets containing six datasets and 795 unlabeled samples. Results showed that the proposed model directly transferred to the different datasets without labeled samples, and achieved average buffered Hausdorff scores of 90.17–96.42, recall scores of 0.84–0.91, and F-values of 0.85–0.89. The PCDM-HED demonstrated a strong TL capability across heterogeneous 3D and 2D crack datasets.

The robustness of the PCDM-HED is based on preprocessed edge enhancement and deep edge characteristics using multilayer networks. The fusion of enhanced edge features and data-driven methods helps balance the accuracy and generalization. However, the current PCDM-HED has three shortcomings. First, pavement non-crack objects with edges are detected, such as joints in cement pavements, grooves in airport pavements, and curbstone edges. To some extent, these misidentifications are detrimental to pavement maintenance decision-making. Second, the crack severities were obtained through post-processing, and future studies can focus on using more accurate crack-severity labels to directly obtain crack location and severity using networks. Third, the robustness of the method is based on pre-processing and networks rather than on end-to-end methods, and to some extent, the degree of automation is not as good as that of end-to-end networks. In summary, future work should focus on further improving the network model and refining the labeled crack-severity datasets to better integrate multiple edge features and improve the degree of method automation.

CRediT authorship contribution statement

Rong Gui: Conceptualization, Data curation, Methodology, Visualization, Writing – original draft, Writing – review and editing. **Qian Sun:** Conceptualization, Methodology, Visualization, Writing – original draft, Writing – review and editing. **Wenqing Wu:** Data curation, Methodology, Visualization, Writing – original draft. **Dejin Zhang:** Conceptualization, Writing – review and editing. **Qingquan Li:** Conceptualization, Writing – review and editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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